Phenix Cryo-EM Tools (Part 1 of 2)

11 documentation pages
Generated from local Phenix documentation

Contents

Docking a model into a map

cryo-em-dock-model.html

Why

Obtaining the structure of a macromolecule generally requires building an atomic model that fits the cryo-EM map. A docking procedure is used to do this fitting if the model, or a part of the model, is known but has not been placed into the map.

For example, your cryo-EM map might show a molecular complex assembled from components available from other experiments, such as X-ray crystallography. Then, you can dock these components' models into your cryo-EM map to obtain a model for the entire complex.

How

The program phenix.dock_in_map automatically docks one or several models into a map, using a convolution-based shape search to find a part of the map that is similar to the model. Initially, it performs a low-resolution search that focuses on the overall shape of the molecule, which is performed without rotation to optimize runtime. Alternatively, you can perform an initial fast search by matching the moments of inertia of the model and map, but this option requires an accurate map that looks a lot like the model. Next, if the placement is satisfactory, a full search can be run on multiple processors using real-space rigid-body refinement with the full resolution of the map.

In a standard run of phenix.dock_in_map, you input the model (PDB file) that you want to place, a map (CCP4, MRC, or other map format file), the nominal resolution of the map, and the number of processors to use.The program outputs a search PDB file transformed to superpose on the target map file

Getting help from the Phenix chatbot

You can use the free Phenix chatbot to get help on any Phenix program. Just ask it any question about using Phenix.

How to use phenix.dock_in_map: Click here

Common issues

- None

Related programs

- cryo_fit: This tool performs flexible fitting of a model into a cryo-EM map.

Map improvement (cryo-EM)

Why

Cryo-EM maps can be improved by methods such as sharpening/blurring and density modification.

Sharpening can reveal the high-resolution details concealed in the cryo-EM map. Often, cryo-EM maps can appear smooth and can lack a high level of detail (contrast) because high-resolution amplitudes of corresponding Fourier map coefficients decay from causes such as radiation damage, sample movement, sample heterogeneity, and errors in the reconstruction procedure. We note that sharpening (and blurring) modify the amplitudes of Fourier coefficients and leave the phases unchanged.

Density modification can be used to improve a cryo-EM map by adjusting the Fourier coefficients to better agree with both the original map and the expected features. This improvement relies on two ideas: (1) the Fourier coefficients representing the map are uncorrelated, and (2) some features in a map region are known in advance.

How

The program phenix.map_sharpening performs map sharpening by optimizing the detail and connectivity of a cryo-EM map. It tests several resolution-dependent functions and outputs the one with maximal clarity, based on adjusted surface area (default) or kurtosis. The program works best on maps with 4.5 Å or better resolution. At lower resolutions, it may be unable to distinguish variations in the quality of the map as the sharpening is varied. Typically, optimal sharpening is determined by examining a box cut out of the map, and then applying this to the entire map; however, other options are available.

The program phenix.resolve_cryo_em performs density modification of a cryo-EM map using two unmasked half-maps, the FSC-based resolution, and a sequence file specifying the contents of the map. You can also supply a model, which will be randomized, refined, and used in model-based density modification. You normally access the program's functionality by running the ResolveCryoEM tool in the Phenix GUI.

How to use phenix.map_sharpening: Click here

How to use phenix.resolve_cryo_em: Click here

Common issues

- phenix.map_sharpening doesn't work for map_box-produced maps: map_sharpening should not be used on maps produced with the exact-unique option of map_box. These maps are closely masked around the density of a single molecular, so much of the map is set to zero. Therefore, the map is missing noise information and auto-sharpen doesn't work properly.
- phenix.resolve_cryo_em doesn't work well: Poor results could be due to using masked half-maps, having non-uniform solvent noise, or getting interference from having very prominent density away from the macromolecule. The program can also crash if you have limited memory and are using multiple processors. See phenix.resolve_cryo_em for ways to address these issues.

Related programs

- phenix.map_symmetry: This tool finds internal symmetries in a map, allowing the map to be reduced to the repeating unit.
- phenix.map_box: This tool cuts out a box from a larger map. It is used to extract the unique part of a map.

- phenix.combine_focused_maps: This tool creates a weighted composition map from a set of locally focused maps and associated models, where each part of each map is weighted by its correlation with the corresponding model. The maps do not have to be superimposed or even have the same gridding or size.

Assessing Map Quality (cryo-EM)

cryo-em-map-quality.html

Why

The first step after data processing is analysis to assess data quality and detect any pathologies that might thwart structure solution. The quality of the reconstruction, and therefore the interpretability of a cryo-EM map, can deteriorate from many causes including structural heterogeneity, radiation damage, and beam-induced sample movement.

Data quality is typically expressed by the resolution, but this term has a different meaning for cryo-EM than crystallographic experiments. For a cryo-EM map, the overall resolution (dFSC) is usually defined as the maximum spatial resolution at which the information content of the map is reliable.

The cryo-EM resolution is typically obtained by analyzing the Fourier shell correlation (FSC) for two cryo-EM half-maps binned in resolution shells. If the macromolecule is structurally heterogeneous, a single resolution value is adequate. 'Local resolutions' can be also assigned to different map regions.

How

The phenix.mtriage program estimates the resolution of cryo-EM maps using several approaches, and then provides a summary of the map statistics. The tool requires only a full cryo-EM map (in CCP4, MRC, or other related format). However, you can obtain additional information by inputting two half-maps and/or an atomic model (in PDB or mmCIF format).

How to use the phenix.mtriage: Click here

Common issues

Program stops: All input maps are expected to be on the same grid. Also, box information (e.g., unit cell of CRYST1 record in PDB files) must match across all of the inputs.

Related programs and Documentation

- phenix.show_map_info: This program lists the properties of a map, including the origin, grid points, unit cell, and map size.
- phenix.map_box: This program cuts out a box around selected atoms from a larger map (MTZ, CCP4, or X-plor format).
- phenix.map_symmetry: Some molecules (e.g., viruses) have high internal symmetry, so you can reduce the map to the repeating unit. This program finds such symmetries, and phenix.map_box can extract the unique part of the map (using the extract-unique option).
- phenix.combine_focused_maps: This program creates a weighted composite map from a set of locally focused maps and associated models, where each part of each map is weighted by its correlation with the corresponding model.

Map Symmetry (cryo-EM)

cryo-em-map-symmetry.html

Why

Some molecules, such as viruses, have high internal symmetry so you can reduce the cryo-EM map to the repeating unit. This can decrease the required computational resources and enable better averaging for improved signal to noise.

How

The phenix.map_symmetry program uses the density at a set of randomly-chosen points in a cryo-EM map to identify symmetries, which are selected from a set of pre-defined symmetry types (e.g., point-group symmetry and helical symmetry). The tool then creates a file with the symmetry operations.

By default, phenix.map_symmetry assumes the principle axes of symmetry in a map are along x, y, and z and that the helical symmetry is along z. The user can specify a specific symmetry type or look for all elements in the database of 45 common symmetry elements and point groups.

How to use the phenix.map_symmetry: Click here

Common issues

None.

Related programs and Documentation

- phenix.map_box: This program cuts out a box around selected atoms from a larger map. It can be used to extract the unique part of the map (using the extract-unique option).
- phenix.combine_focused_maps: This program creates a weighted composite map from a set of locally focused maps and associated models, where each part of each map is weighted by its correlation with the corresponding model.

Automated model building (cryo-EM)

cryo-em-model-building.html

Why

Determining the structure of a macromolecule typically requires building an atomic model that fits the cryo-EM map. If the structure of the molecule or its components are unknown, the model has to be built from scratch. This task is challenging because molecules are typically very large and map interpretation is difficult at low resolution. Therefore, manual interpretation of cryo-EM maps is time-consuming and error-prone. For many cases, however, automated methods can be used to build models for both proteins and nucleic acids.

How

The program phenix.map_to_model interprets a cryo-EM map and automatically builds an atomic model. If desired, the map is sharpened using phenix.map_sharpening. The unique parts of the structure are then identified by taking the reconstruction symmetry into account. The procedure identifies which parts of the map correspond to protein or RNA/DNA. To save runtime, you can use phenix.map_sharpening to sharpen the map and phenix.map_box to cut out the unique part before you run phenix.map_to_model.

The phenix.map_to_model tool requires you to input a CCP4-style (MRC, etc) cryo-EM map or MTZ map coefficients and a sequence file. For optimal model-building results, the resolution of the map should be 4.5 Å or better. The program outputs a PDB file with a model that is superimposed on the original map.

How to use the phenix.map_to_model GUI: Click here

Common issues

- One or more sub-processes crash: If you have a very large structure, your computer may not have enough memory to run phenix.map_to_model. The easiest solution is to run the tool on just the unique part of the structure. You can also cut out boxes and work on each section individually. Or you can cut back on the resolution to save memory.
- Want to re-run without all the steps: Partially-completed jobs can occur when your queueing system crashes during a run or one or more sub-processes crash. If you have a partially-completed or completed run, you can carry on from where you left off by using the keyword carry_on=True. You can also use this keyword to build up your model in pieces or to run many jobs in parallel.

Related programs

- phenix.map_sharpening: This program performs map sharpening by optimizing the detail and connectivity of a cryo-EM map.
- phenix.map_box: This program cuts out a box around selected atoms from a larger map (MTZ, CCP4, or X-plor format).
- phenix.fix_insertions_deletions: This program fixes register errors in the model by comparing the density of side-chain positions to a sequence file.
- phenix.trace_and_build: This tool rapidly builds protein models by tracing the path of a polypeptide chain, finding the CB positions, and building the atomic model. Normally, this is done by running phenix.map_to_model, but you can also run phenix.trace_and_build directly.

Real-space refinement (cryo-EM)

cryo-em-real-space-refinement.html

Why

Once an atomic model has been built using a cryo-EM map, you need to optimize it to best fit the experimental data while also preserving good agreement with prior chemical knowledge. This process alternates between automated refinement, validation, and either manual or automated model corrections.

A target function guides the refinement by linking the model parameters to the experimental data and by scoring the model-versus-data fit. For cryo-EM data, refinement of the model is done in real space and the target function is formulated in terms of a three-dimensional map. Because there are generally too many model parameters, refinement requires additional restraints that modify the target function by creating relationships between independent parameters.

How

The program used to refine atomic models in real space is phenix.real_space_refine. The algorithm uses a simplified refinement target function that speeds up calculations, so optimal data-restraint weights can be identified with little runtime cost. The procedure is robust, working at resolutions from 1 to 6 Å. The input and output models can be in PDB or mmCIF formats. The input map can be in CCP4 or MTZ format.

This program makes use of restraints on covalent geometry, secondary structure, Ramachandran plots, rotamer outliers, and internal molecular symmetry. The default mode performs gradient-driven minimization of the entire model. However, optimization can also be performed using simulated annealing, morphing, rigid-body refinement, and/or systematic side-chain improvement — at the potential expense of longer runtimes.

How to use phenix.real_space_refine: Click here

Common issues

- Poorly refined model: If you use secondary structure restraints, an incorrectly refined model is likely if there are errors in the secondary structure annotation (e.g., incorrect SHEET and HELIX records in the PDB file header). Secondary structure annotation strongly depends on the quality of the input model.

Cryo-EM Structure Deposition

cryo-em-structure-deposition.html

Why

Once a protein structure is solved, it is usually deposited in the World Wide Protein Data Bank (wwPDB). This is necessary for publication as most journals require a PDB ID to be included in any manuscript that describes a new crystallographic structure. The wwPDB is an invaluable resource that contains over 190,000 structures (as of July 2021).

Procedure

The model files (PDB and mmCIF) generated by phenix.real_space_refine contain information that is required when depositing a structure to the wwPDB. However, the wwPDB currently only accepts model files in mmCIF for deposition.

To generate a model file suitable to deposition, a two stage process is currently recommended:

- By default, phenix.real_space_refine will output model files in mmCIF and PDB format. If you turned off the option for mmCIF output, run a final cycle of phenix.real_space_refine that writes mmCIF files for model and data. This can be set in the Output section of the GUI.
- You can then process the model file (mmCIF) and a sequence file with the mmtbx.prepare_pdb_deposition program to create a mmCIF file with the sequence. This program requires the full sequence for the macromolecule to be provided. In the GUI, this program is in the "PDB Deposition" section of tools.

How to use mmtbx.prepare_pdb_deposition: Click here

Related programs

- phenix.validation_cryoem: This program can be used to generate the table of statistics.
- phenix.get_pdb_validation_report: This program submits your model and data in CIF format to the PDB OneDep server to get a validation report in PDF an XML formats.

References

Announcing mandatory submission of PDBx/mmCIF format files for crystallographic depositions to the Protein Data Bank (PDB). P.D. Adams, P.V. Afonine, K. Baskaran, H.M. Berman, J. Berrisford, G. Bricogne, D.G. Brown, S.K. Burley, M. Chen, Z. Feng, C. Flensburg, A. Gutmanas, J.C. Hoch, Y. Ikegawa, Y. Kengaku, E. Krissinel, G. Kurisu, Y. Liang, D. Liebschner, L. Mak, J.L. Markley, N.W. Moriarty, G.N. Murshudov, M. Noble, E. Peisach, I. Persikova, B.K. Poon, O.V. Sobolev, E.L. Ulrich, S. Velankar, C. Vonrhein, J. Westboork, M. Wojdyr, M. Yokochi, and J.Y. Young. Acta Cryst. D75, 451-454 (2019).

- Announcing mandatory submission of PDBx/mmCIF format files for crystallographic depositions to the Protein Data Bank (PDB). P.D. Adams, P.V. Afonine, K. Baskaran, H.M. Berman, J. Berrisford, G. Bricogne, D.G. Brown, S.K. Burley, M. Chen, Z. Feng, C. Flensburg, A. Gutmanas, J.C. Hoch, Y. Ikegawa, Y. Kengaku, E. Krissinel, G. Kurisu, Y. Liang, D. Liebschner, L. Mak, J.L. Markley, N.W. Moriarty, G.N. Murshudov, M. Noble, E. Peisach, I. Persikova, B.K. Poon, O.V. Sobolev, E.L. Ulrich, S. Velankar, C. Vonrhein, J. Westboork, M. Wojdyr, M. Yokochi, and J.Y. Young. Acta Cryst. D75, 451-454 (2019).

Validation (cryo-EM)

cryo-em-validation.html

Why

Comprehensive Validation (cryo-EM) is a tool that can be used to evaluate the plausibility of a model, including its agreement with cryo-EM data. Note that for crystallographic (X-ray or neutron) data you should use instead Comprehensive Validation (X-ray).

Many steps are involved to get from the initial idea about a molecule to the final atomic model and its interpretation. Errors can occur in each of these steps, so validation procedures are necessary to ensure that the atomic models are reasonable and fit the experimental data.

In particular, validation addresses data, model, and model-versus-data quality:

• Data. The quality of experimental data, such as the resolution or amount of noise, defines the quality of the derived models. Therefore, it is vital to understand the data quality so that you use the correct model-building and refinement procedures appropriate for the data at hand. (See the Assessing Map Quality (cryo-EM) overview for more details.)

• Model. The underlying principles behind model validation are the same for any experimental method: a good model should make chemical sense and be consistent with empirical statistics for high-quality prior structures. Thus, atomic models derived from cryo-EM reconstructions should conform to prior knowledge about covalent geometry, non-bonded interactions, and known distributions of side-chain and main-chain conformations in both proteins and nucleic acids. Generally, the goal is to have as few outliers as is feasible, where outliers can be explained by their environment and/or experimental data.

• Model-versus-data fit. Model-verses-data validation criteria require the model to describe its own data well. A perfectly good model from a stereochemistry perspective may not fit the 3D reconstruction sufficiently or at all. In reciprocal space, model-versus-data agreement is assessed by curves of the Fourier shell correlation (FSC) as a function of resolution. In real space, the agreement is measured by correlation coefficients (CCs).

How

Comprehensive validation (cryo-EM) is a one-stop program in Phenix that validates model, data, and model-versus-data fit. The minimal input is a map (in MRC, CCP4, or related format) and an atomic model (in PDB or mmCIF format). However, providing a map, model, and two half-maps is desirable.

For geometric validation, Phenix uses MolProbity, which is fully integrated into the Comprehensive validation (cryo-EM) package. The graphics programs Coot and PyMOL are also fully integrated, enabling communication of the validation results.

How to use the validation tools in the Phenix GUI: Click here

Common issues

• Outliers versus errors: In most structures, there will be Ramachandran and side-chain rotamer outliers. If the data (the map) and surrounding chemistry supports the outlier conformation, then it is most likely an outlier from prior expectations. In the absence of supporting density and/or conflicting local chemistry, it is probably an error that needs correction. For example, an atomic model derived from low-resolution data is expected to have no geometrical outliers because they are unlikely to be supported by the experimental data.

Related programs

- phenix.cablam_validation: The C-Alpha Based Low-Resolution Annotation Method (CaBLAM) validates protein backbone conformations in models determined at low resolution (2.5-4 Å), where it is difficult to determine peptide orientations because carbonyl O atoms cannot be discerned in the density maps. The program uses protein C-alpha geometry to evaluate the main-chain geometry and identify areas of probable secondary structure.
- phenix.mtriage: This program computes various cryo-EM map statistics, including resolution estimates, Fourier shell correlation curves, and more.

Cryo-EM structure solution with Phenix

cryo-em_index.html

Click on boxes in the figure to read an overview of each step. Also see the AlphaFold models in Phenix overview and video tutorials for using AlphaFold models in Phenix. You can use the Phenix chatbot to get help on using any Phenix program. Just ask it any question about using Phenix. GUI documentation for most-used programs: Predict a structure with AlphaFold using PredictModel Automated structure determination using AlphaFold models with PredictAndBuild Assess the quality of your map with Mtriage Optimize a map with MapSharpening Density modify a map with RESOLVE cryo-EM Identify symmetry in a map with Map-Symmetry Extract a box with map and model with Map-Box Dock a model into a map with Dock-in-map Flexibly fit a model to a map with CryoFit Build a model with Map-to-model Refine your model against a map with phenix.real_space_refine Generate geometry restraints Build ordered water with Douse Model heterogenicity by refinement against multiple maps phenix.varref Complete list of programs for each structure solution step Other map tools: Calculate a local resolution map Combine best parts of focused maps Convert map to structure factors Segment a map Other model building tools: Rapid model-building Guess sequences from map Compare CA/P in two models Adjust a CA/CB model Fix register errors

GUI documentation for most-used programs:

- Predict a structure with AlphaFold using PredictModel
- Automated structure determination using AlphaFold models with PredictAndBuild
- Assess the quality of your map with Mtriage
- Optimize a map with MapSharpening
- Density modify a map with RESOLVE cryo-EM
- Identify symmetry in a map with Map-Symmetry
- Extract a box with map and model with Map-Box
- Dock a model into a map with Dock-in-map
- Flexibly fit a model to a map with CryoFit
- Build a model with Map-to-model
- Refine your model against a map with phenix.real_space_refine
- Generate geometry restraints
- Build ordered water with Douse
- Model heterogenicity by refinement against multiple maps phenix.varref
- Complete list of programs for each structure solution step

Other map tools: Calculate a local resolution map Combine best parts of focused maps Convert map to structure factors Segment a map

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Auto-sharpen (deprecated, use MapSharpening)

auto_sharpen.html

Author(s)

- auto_sharpen: Tom Terwilliger

Purpose

The routine auto_sharpen will automatically identify optimal sharpening/blurring/map adjustment for the input map and will write out an optimized version of the map.

Video Tutorial

The tutorial video is available on the Phenix YouTube channel and covers the following topics:

- basic overview
- how to run phenix.map_to_model via the GUI

GUI

A Graphical User Interface is available.

Usage

How auto_sharpen works:

Auto-sharpen adjusts the resolution dependence of the map to maximize the clarity of the map. You can choose to use map kurtosis or the adjusted surface area of the map which is the default for this purpose.

The auto-sharpen tool works best at resolutions of about 4.5 A or better. At lower resolutions it may be unable to distinguish variations in the quality of the map as the sharpening is varied.

Kurtosis is a standard statistical measure that reflects the peakiness of the map.

The adjusted surface area is a combination of the surface area of contours in the map at a particular threshold and of the number of distinct regions enclosed by the top 30% (default) of those contours. The threshold is chosen by default to be one where the volume enclosed by the contours is 20% of the non-solvent volume in the map. The weighting between the surface area (to be maximized) and number of regions enclosed (to be minimized) is chosen empirically (default region_weight=20).

Several resolution-dependent functions are tested, and the one that gives the best adjusted surface area (or kurtosis) is chosen. In each case the map is transformed to obtain Fourier coefficients. The amplitudes of these coefficients are then adjusted, keeping the phases constant. The available functions for modifying the amplitudes are:

```
No sharpening (map is left as is)

Sharpening b-factor applied over entire resolution range (b_sharpen
applied to achieve an effective isotropic overall b-value of b_iso).

Sharpening b-factor applied up to resolution specified with the
resolution=xxx keyword, then blurred beyond this resolution (with
```

```
transition specified by the keyword k_sharpen, b_iso_to_d_cut). If
the sharpening b_sharpen is negative (blurring the map),
the blurring is applied over the entire resolution range.

Resolution-dependent sharpening factor with three parameters.
First the resolution-dependence of the map is removed by normalizing the
amplitudes. Then a scale factor S is to the data, where
log10(S) is determined by coefficients b[0],b[1],b[2] and a resolution
d_cut (typically d_cut is the nominal resolution of the map).
The value of log10(S) varies smoothly from 0 at resolution=infinity, to b[0]
at d_cut/2, to b[1] at d_cut, and to b[1]+b[2] at the highest resolution
in the map. The value of b[1] is limited to being no larger than b[0] and the
value of b[1]+b[2] is limited to be no larger than b[1].
```

You can also choose to specify the sharpening/blurring parameters for your map and they will simply be applied to the map. For example you can apply a sharpening B-value (b_sharpen) to sharpen the map, or you can specify a target overall B-value (b_iso) to obtain after sharpening.

Box of density in sharpening

Normally phenix.auto_sharpen will determine the optimal sharpening by examining the density in a box cut out of your map, then apply this to the entire map.

Local sharpening

You can choose to apply autosharpening locally if you want. In this case the auto-sharpening parameters are determined in many boxes cut out of the map, and corresponding sharpened maps are calculated. The map that is produced is a weighted map where the density at a particular point comes most from the sharpened map based on a box near that point.

Half-map-based sharpening

You can identify the sharpening parameters using two half-maps if you want. The resolution-dependent correlation of density in the two half-maps is used to identify the optimal resolution-dependent weighting of the map. This approach requires a target resolution which is used to set the overall fall-off with resolution for an ideal map. That fall-off for an ideal map is then multiplied by an estimated resolution-dependent correlation of density in the map with the true map (the estimation comes from the half-map correlations).

Model-based sharpening

You can identify instead the sharpening parameters using your map and a model. This approach requires a guess of the RMSD between the model and the true model. The resolution-dependent correlation of model and map density is used as in the half-map approach above to identify the weighting of Fourier coefficients.

Using crystallographic maps

You can use phenix.auto_sharpen with a crystallographic map (represented as map coefficients).

Shifting the map to the origin

Most crystallographic maps have the origin at the corner of the map (grid point [0,0,0]), while most cryo-EM maps have the orgin in the middle of the map. An output map with the origin shifted to the corner of the map is optionally written out.

Output files from auto_sharpen

sharpened_map.ccp4: Sharpened map.

shifted_sharpened_map.ccp4: Sharpened map, shifted to place the origin on grid point (0,0,0) and sharpened

sharpened_map_coeffs.mtz: Sharpened map, shifted to place the origin on grid point (0,0,0) and sharpened, represented as map coefficients.

Examples

Standard run of auto_sharpen:

Running auto_sharpen is easy. From the command-line you can type:

```
phenix.auto_sharpen my_map.map resolution=2.6
```

where my_map.map is a CCP4, mrc or other related map format, and you specify the nominal resolution of the map.

Possible Problems

Specific limitations and problems:

Maps produced with the extract-unique option of map_box should not be sharpened with auto-sharpen. These maps are closely masked around the density of a single molecule and are set to zero in much of the map, so the information about noise in the map that normally is available is missing and auto-sharpen does not work properly.

Literature

Automated map sharpening by maximization of detail and connectivity. T.C. Terwilliger, P.V. Afonine, Sobolev, OV, and P.D. Adams. Acta Cryst. D74, 545-559 (2018).

- Automated map sharpening by maximization of detail and connectivity. T.C. Terwilliger, P.V. Afonine, Sobolev, OV, and P.D. Adams. Acta Cryst. D74, 545-559 (2018).

Additional information

List of all available keywords

job_title = None Job title in PHENIX GUI, not used on command lineinput_filesmap_file = None File with CCP4-style maphalf_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_fileexternal_map_file = None External map to be used to scale map_file (power vs resolution will be matched)map_coeffs_file = None Optional file with map coefficientsmap_coeffs_labels = None Optional label specifying which columns of of map coefficients to usepdb_file = None If a model is supplied, the map will be adjusted to maximize map-model correlation. This can be used to improve a map in regions where no model is yet built.ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.seq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file.input_weight_map_pickle_file = None Weight map pickle fileoutput_filesshifted_map_file =

shifted_map.ccp4 Input map file shifted to new origin.sharpened_map_file = None Sharpened input map file. In the same location as input map.shifted_sharpened_map_file = None Input map file shifted to place origin at 0,0,0 and sharpened.sharpened_map_coeffs_file = None Sharpened input map (shifted to new origin if original origin was not 0,0,0), written out as map coefficientsoutput_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle fileoutput_directory = None Directory where output files are to be written applied.crystal_infois_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry=True and False otherwise.resolution = None Optional nominal resolution of the map.solvent_content = None Optional solvent fraction of the cell.solvent_content_iterations = 3 Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.ncs_copies = None You can specify ncs copies and seq file to define solvent contentwang_radius = None Wang radius for solvent identification. Default is 1.5* resolutionbuffer_radius = None Buffer radius for mask smoothing. Default is resolutionpseudo_likelihood = None Use pseudo-likelihood method for half-map sharpening. (In development)map_modificationb_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_isob_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolutionnormalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpeningd_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this valueinput_d_cut = None High-resolution limit for sharpeningrmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.rmsd_resolution_factor ...

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- input_filesmap_file = None File with CCP4-style maphalf_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_fileexternal_map_file = None External map to be used to scale map_file (power vs resolution will be matched)map_coeffs_file = None Optional file with map coefficientsmap_coeffs_labels = None Optional label specifying which columns of of map coefficients to usepdb_file = None If a model is supplied, the map will be adjusted to maximize map-model correlation. This can be used to improve a map in regions where no model is yet built.ncs_file = None File with NCS information (typically point-group NCS with the center specified). Typically in PDB format. Can also be a .ncs_spec file from phenix. Created automatically if symmetry is specified.seq_file = None Sequence file (unique chains only, 1-letter code, chains separated by blank line or greater-than sign.) Can have chains that are DNA/RNA/protein and all can be present in one file.input_weight_map_pickle_file = None Weight map pickle file
- map_file = None File with CCP4-style map
- half_map_file = None Half map (two should be supplied) for FSC calculation. Must have grid identical to map_file
- external_map_file = None External map to be used to scale map_file (power vs resolution will be matched)
- map_coeffs_file = None Optional file with map coefficients

- map_coeffs_labels = None Optional label specifying which columns of of map coefficients to use
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- input_weight_map_pickle_file = None Weight map pickle file
- output_filesshifted_map_file = shifted_map.ccp4 Input map file shifted to new origin.sharpened_map_file = None Sharpened input map file. In the same location as input map.shifted_sharpened_map_file = None Input map file shifted to place origin at 0,0,0 and sharpened.sharpened_map_coeffs_file = None Sharpened input map (shifted to new origin if original origin was not 0,0,0), written out as map coefficientsoutput_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle fileoutput_directory = None Directory where output files are to be written applied.
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- output_weight_map_pickle_file = weight_map_pickle_file.pkl Output weight map pickle file
- output_directory = None Directory where output files are to be written applied.
- crystal_infois_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry=True and False otherwise.resolution = None Optional nominal resolution of the map.solvent_content = None Optional solvent fraction of the cell.solvent_content_iterations = 3 Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.ncs_copies = None You can specify ncs copies and seq file to define solvent contentwang_radius = None Wang radius for solvent identification. Default is 1.5* resolutionbuffer_radius = None Buffer radius for mask smoothing. Default is resolutionpseudo_likelihood = None Use pseudo-likelihood method for half-map sharpening. (In development)
- is_crystal = None Defines whether this is a crystal (or cryo-EM). Default is True if use_sg_symmetry=True and False otherwise.
- resolution = None Optional nominal resolution of the map.
- solvent_content = None Optional solvent fraction of the cell.
- solvent_content_iterations = 3 Iterations of solvent fraction estimation. Used for ID of solvent content in boxed maps.
- molecular_mass = None Molecular mass of molecule in Da. Used as alternative method of specifying solvent content.
- ncs_copies = None You can specify ncs copies and seq file to define solvent content
- wang_radius = None Wang radius for solvent identification. Default is 1.5* resolution

- buffer_radius = None Buffer radius for mask smoothing. Default is resolution
- pseudo_likelihood = None Use pseudo-likelihood method for half-map sharpening. (In development)
- map_modificationb_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_isob_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolutionnormalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpeningd_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this valueinput_d_cut = None High-resolution limit for sharpeningrmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factorfraction_complete = None Completness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).auto_sharpen = True Automatically determine sharpening using kurtosis maximization or adjusted surface area. Default is Trueauto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut external_map_sharpening None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.box_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map. Default is False.density_select_in_auto_sharpen = True Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally use False for X-ray data and True for cryo-EM.density_select_threshold_in_auto_sharpen = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.soft_mask = True Use soft mask (smooth change from inside to outside with radius based on resolution of map).use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)discard_if_worse = None Discard sharpening if worselocal_sharpening = None Sharpen locally using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if NCS is present. If local_aniso_in_local_sharpening is True and NCS is present this can distort the map for some NCS copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless NCS is present.overall_before_local = True Apply overall scaling before local scalingselect_sharpened_map = None Select a single sharpened map to useread_sharpened_maps = No...
- b_iso = None Target B-value for map (sharpening will be applied to yield this value of b_iso). If sharpening method is not supplied, default is to use b_iso_to_d_cut sharpening.
- b_sharpen = None Sharpen with this b-value. Contrast with b_iso that yields a targeted value of b_iso

• b_blur_hires = 200 Blur high_resolution data (higher than d_cut) with this b-value. Contrast with b_sharpen applied to data up to d_cut. Note on defaults: If None and b_sharpen is positive (sharpening) then high-resolution data is left as is (not sharpened). If None and b_sharpen is negative (blurring) high-resolution data is also blurred.

• resolution_dependent_b = None If set, apply resolution_dependent_b (b0 b1 b2). Log10(amplitudes) will start at 1, change to b0 at half of resolution specified, changing linearly, change to b1/2 at resolution specified, and change to b1/2+b2 at d_min_ratio*resolution

• normalize_amplitudes_in_resdep = False Normalize amplitudes in resolution-dependent sharpening

• d_min_ratio = 0.833 Sharpening will be applied using d_min equal to d_min_ratio times resolution. Default is 0.833

• scale_max = 100000 Scale amplitudes from inverse FFT to yield maximum of this value

• input_d_cut = None High-resolution limit for sharpening

• rmsd = None RMSD of model to true model (if supplied). Used to estimate expected fall-of with resolution of correct part of model-based map. If None, assumed to be resolution times rmsd_resolution_factor.

• rmsd_resolution_factor = 0.25 default RMSD is resolution times resolution factor

• fraction_complete = None Completness of model (if supplied). Used to estimate correct part of model-based map. If None, estimated from max(FSC).

• auto_sharpen = True Automatically determine sharpening using kurtosis maximization or adjusted surface area. Default is True

• auto_sharpen_methods = no_sharpening b_iso *b_iso_to_d_cut resolution_dependent model_sharpening half_map_sharpening target_b_iso_to_d_cut external_map_sharpening None Methods to use in sharpening. b_iso searches for b_iso to maximize sharpening target (kurtosis or adjusted_sa). b_iso_to_d_cut applies b_iso only up to resolution specified, with fall-over of k_sharpen. Resolution dependent adjusts 3 parameters to sharpen variably over resolution range. Default is b_iso_to_d_cut . target_b_iso_to_d_cut uses target_b_iso_ratio to set b_iso.

• box_in_auto_sharpen = False Use a representative box of density for initial auto-sharpening instead of the entire map. Default is False.

• density_select_in_auto_sharpen = True Choose representative box of density for initial auto-sharpening with density_select method (choose region where there is high density). Normally use False for X-ray data and True for cryo-EM.

• density_select_threshold_in_auto_sharpen = None Threshold for density select choice of box. Default is 0.05. If your map has low overall contrast you might need to make this bigger such as 0.2.

• allow_box_if_b_iso_set = False Allow box_in_auto_sharpen (if set to True) even if b_iso is set. Default is to set box_n_auto_sharpen=False if b_iso is set.

• soft_mask = True Use soft mask (smooth change from inside to outside with radius based on resolution of map).

• use_weak_density = False When choosing box of representative density, use poor density (to get optimized map for weaker density)

• discard_if_worse = None Discard sharpening if worse

• local_sharpening = None Sharpen locally using overlapping regions. NOTE: Best to turn off local_aniso_in_local_sharpening if NCS is present. If local_aniso_in_local_sharpening is True and NCS is present this can distort the map for some NCS copies because an anisotropy correction is applied based on local density in one copy and is transferred without rotation to other copies.

• local_aniso_in_local_sharpening = None Use local anisotropy in local sharpening. Default is True unless NCS is present.

• overall_before_local = True Apply overall scaling before local scaling

• select_sharpened_map = None Select a single sharpened map to use

• read_sharpened_maps = None Read in previously-calculated sharpened maps

• write_sharpened_maps = None Write out local sharpened maps

• smoothing_radius = None Sharpen locally using smoothing_radius. Default is 2/3 of mean distance between centers for sharpening

• box_center = None You can specify the center of the box (A units)

• box_size = 30 30 30 You can specify the size of the box (grid units)

• target_n_overlap = 10 You can specify the targeted overlap of boxes in local sharpening

• restrict_map_size = None Restrict box map to be inside full map (required for cryo-EM data). Default is True if use_sg_symmetry=False and False if use_sg_symmetry=True

• remove_aniso = True You can remove anisotropy (overall and locally) during sharpening

• max_box_fraction = 0.5 If box is greater than this fraction of entire map, use entire map. Default is 0.5.

• density_select_max_box_fraction = 0.95 If box is greater than this fraction of entire map, use entire map for density_select. Default is 0.95

• cc_cut = 0.2 Estimate of minimum highly reliable CC in half-map FSC. Used to decide at what CC value to smooth the remaining CC values.

• max_cc_for_rescale = 0.2 Used along with cc_cut and scale_using_last to correct for small errors in FSC estimation at high resolution. If the value of FSC near the high-resolution limit is above max_cc_for_rescale, assume these values are correct and do not correct them.

• scale_using_last = 3 If set, assume that the last scale_using_last bins in the FSC for half-map or model sharpening are about zero (corrects for errors int the half-map process).

• mask_atoms = True Mask atoms when using model sharpening

• mask_atoms_atom_radius = 3 Mask for mask_atoms will have mask_atoms_atom_radius

• value_outside_atoms = None Value of map outside atoms (set to mean to have mean value inside and outside mask be equal)

• k_sharpen = 10 Steepness of transition between sharpening (up to resolution) and not sharpening (d < resolution). Note: for blurring, all data are blurred (regardless of resolution), while for sharpening, only data with d about resolution or lower are sharpened. This prevents making very high-resolution data too strong. Note 2: if k_sharpen is zero or None, then no transition is applied and all data is sharpened or blurred.

• iterate = False You can iterate auto-sharpening. This is useful in cases where you do not specify the solvent content and it is not accurately estimated until sharpening is optimized.

• optimize_b_blur_hires = False Optimize value of b_blur_hires. Only applies for auto_sharpen_methods b_iso_to_d_cut and b_iso. This is normally carried out and helps prevent over-blurring at high resolution if the same map is sharpened more than once.

• optimize_d_cut = None Optimize value of d_cut. Only applies for auto_sharpen_methods b_iso_to_d_cut and b_iso

- adjust_region_weight = True Adjust region_weight to make overall change in surface area equal to overall change in normalized regions over the range of search_b_min to search_b_max using b_iso_to_d_cut.
- region_weight_method = initial_ratio *delta_ratio b_iso Method for choosing region_weights. Initial_ratio uses ratio of surface area to regions at low B value. Delta ratio uses change in this ratio from low to high B. B_iso uses resolution-dependent b_iso (not weights) with the formula b_iso=5.9*d_min**2
- region_weight_factor = 1.0 Multiplies region_weight after calculation with region_weight_method above
- region_weight_buffer = 0.1 Region_weight adjusted to be region_weight_buffer away from minimum or maximum values
- target_b_iso_ratio = 5.9 Target b_iso ratio : b_iso is estimated as target_b_iso_ratio * resolution**2
- target_b_iso_model_scale = 0. For model sharpening, the target_biso is scaled (normally zero).
- signal_min = 3.0 Minimum signal in estimation of optimal b_iso. If not achieved, use any other method chosen.
- search_b_min = None Low bound for b_iso search. Default is -100.
- search_b_max = None High bound for b_iso search. Default is 300.
- search_b_n = None Number of b_iso values to search. Default is 21.
- residual_target = None Target for maximization steps in sharpening. Can be kurtosis or adjusted_sa (adjusted surface area). Default is adjusted_sa.
- sharpening_target = None Overall target for sharpening. Can be kurtosis or adjusted_sa (adjusted surface area). Used to decide which sharpening approach is used. Note that during optimization, residual_target is used (they can be the same.) Default is adjusted_sa.
- require_improvement = None Require improvement in score for sharpening to be applied. Default is True.
- region_weight = None Region weighting in adjusted surface area calculation. Score is surface area minus region_weight times number of regions. Default is set automatically. A smaller value will give more sharpening.
- sa_percent = None Percent of target regions used in calulation of adjusted surface area. Default is 30.
- fraction_occupied = None Fraction of molecular volume targeted to be inside contours. Used to set contour level. Default is 0.20
- n_bins = None Number of resolution bins for sharpening. Default is 20.
- regions_to_keep = None You can specify a limit to the number of regions to keep when generating the asymmetric unit of density.
- max_regions_to_test = None Number of regions to test for surface area in adjusted_sa scoring of sharpening. Default is 30
- eps = None
- k_sol = 0.35 k_sol value for model map calculation. IGNORED (Not applied)
- b_sol = 50 b_sol value for model map calculation. IGNORED (Not applied)
- controlverbose = False Verbose outputresolve_size = None Size of resolve to use. ignore_map_limitations = None Ignore limitations such as map cannot be sharpenedmultiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systemsqueue_run_command = None run command for queue jobs. For example qsub.nproc = 1 Number of processors to use

- verbose = False Verbose output
- resolve_size = None Size of resolve to use.
- ignore_map_limitations = None Ignore limitations such as map cannot be sharpened
- multiprocessing = *multiprocessing sge lsf pbs condor pbspro slurm Choices are multiprocessing (single machine) or queuing systems
- queue_run_command = None run command for queue jobs. For example qsub.
- nproc = 1 Number of processors to use
- guiGUI-specific parameter required for output directoryoutput_dir = None
- output_dir = None

Density modification with with phenix.density_modification

density_modification.html

Density modification with with phenix.density_modification

Contents

Author(s)

- phenix.density_modification: Tom Terwilliger

Purpose

phenix.density_modification is a tool to run Resolve to carry out density modification, including the use of NCS symmetry and electron-density distributions. Masks for the solvent boundary and for the region where NCS operators are to be applied can be specified as PDB files with dummy atoms marking the masked regions.

Usage

How phenix.density_modification works

In phenix.density_modification phases are improved by combining any existing phase information with phase probabilities based on the agreement of your electron density map with expectations about that map. This procedure is known as statisical density modification and is carried out by phenix.resolve. The expectations about a map include:

- A flat solvent region
- Matching NCS-related density
- Density distributions (histograms of density) matching expected distributions
- Density in a region where a model has been built matching expected density for the model

The inputs to phenix.density_modification are :

Required:

- F, SIGF: Structure factor amplitudes for your structure.

Optional:

- PHIB, FOM, HLA HLB HLC HLC: Phases and figure of merit and Hendrickson Lattman (HL) coefficients. These normally come from experimental phasing, but they can be from any source. These are optional if you supply a model. The HL coefficients describe the phase probabilities for each phase.
- Model: This is normally a PDB file containing your current model. It can be used as a source of phase information (calculated starting phases) and also as a target for density modification.
- NCS operators: These are supplied as a .ncs_spec file that you can create by running phenix.find_ncs on a model with NCS or using heavy-atom sites and a map or just with a map.
- Solvent boundary file: This is a PDB file that defines the region that is not solvent (the macromolecule) as dummy atoms.
- NCS boundary file: This is a PDB file that defines the region where NCS is to be applied (all the NCS copies) as dummy atoms.

Density modification starting with phases and HL coefficients

This kind of density modification is normally carried out after experimental phasing, but it can also be carried out if you have created HL coefficients from your data and a model.

In these cases you have phases and phase probability information (F, SIGF, PHIB FOM, HLA HLB HLC HLD).

Prior to density modification, the amplitudes (F, SIGF) are normally corrected for anisotropy and sharpened. You can control this by specifying the target overall B value after sharpening. By default the target overall (Wilson) B is 10 times the high-resolution limit of the data (or the resolution that you specify), up to a maximum of 40 (you can change that too).

A starting map is calculated from F, PHIB, FOM. The location of the macromolecule and solvent are estimated in a probabilistic way using the local variation in the map (with a smoothing radius of rad_wang) to discriminate between the two.

Phases are optimized to agree with the HL coefficients and to yield a map that has a flat solvent, NCS (if present), and density distributions matching model maps.

Density modification with NCS

If you supply NCS operators, phenix.density_modification will automatically figure out the region over which this NCS is present. Within this region, it will weight the density for NCS calculations according to how similar the density is at NCS-related locations. If you supply an NCS mask, that is used instead. The density at NCS-related locations is used as part of the target for density modification.

Density modification with a model

If you supply a model, density modification occurs in several steps. First the data are corrected for anisotropy as above. Then the model is used to estimate phases, fom, and HL coefficients. Then density modification is carried out much as described above for experimental phasing. Then a final step is carried out in which model density is used as part of the target for density modification.

To instead create HL coefficients from your data and model and use those with phenix.density_modification, first run phenix.refine to refine your model. Then run phenix.reciprocal_space_arrays with your model and data and it will create HL coefficients for you.

Note on R-values in density modification

The R-value for density modification is quite different from a refinement R. Basically the statistical (maximum-likelihood) density modification procedure allows calculation of each structure factor based on the values of all the other structure factors and the expectations about the map (i.e, flat solvent, distribution of density values in protein, ncs, etc.) These "map-phasing" structure factors are (at least in the first cycle) independent of the starting structure factors and can be compared with the measured data and an R-value obtained. The values of these R-values range from about 0.25 to 0.5 in most cases, and when the map is very good usually the R values are smaller. These R values do not involve any atomic model so they are quite unlike a refinement R.

For discussion of how map-likelihood structure factors are calculated in Resolve, see the Map-likelihood phasing reference below.

Output files from phenix.density_modification

denmod.mtz: Density-modified map coefficients. If you specify

hl_in_output_mtz=True then it will also contain Hendrickson Lattman coefficients

solvent_boundary.map: CCP4-style map file showing the solvent boundary

ncs_boundary.map: CCP4-style map file showing the asymmetric unit of NCS

Examples

Standard run of phenix.density_modification

Running phenix.density_modification from the command line is easy.

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.5
```

In this example, phaser_1.mtz (the output of Phaser experimental phasing) normally has F, SIGF, PHIB, FOM, HLA HLB HLC HLD, each of which is identified automatically. Standard density modification using these experimental phases and phase probabilities is carried out.

Run specifying resolution, mask type, and output mtz

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.64 \
resolution=3 output_mtz=denmod_std.mtz mask_type=histograms
```

In this example, the high-resolution limit is set to 3 A, the output file is denmod_std.mtz, and the solvent mask is identified using a histogram-based method (mask_type=histograms).

Run specifying a mask for the solvent boundary

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.64 \
mask_from_pdb=mask.pdb rad_mask=4
```

In this example, the file mask.pdb has dummy atoms that indicate where the macromolecule is located. All points within 4 A (rad_mask=4) of an atom in mask.pdb are considered to be within the macromolecule region.

Run specifying NCS operators

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.64 \
ncs_file=ncs.ncs_spec
```

In this example, the file ncs.ncs_spec (created with phenix.find_ncs or phenix.autosol or phenix.simple_ncs_from_pdb or phenix.autobuild) contains the NCS operators. These will be used to automatically find the region where NCS applies. Then the NCS-related density will be used as part of density modification.

Run specifying NCS operators and the region where they are to apply

```
phenix.phenix.density_modification phaser_1.mtz solvent_content=0.64 \
ncs_file=ncs.ncs_spec ncs_domain_pdb=ncs_mask.pdb rad_mask=5
```

This example is like the previous one, except instead of finding the region where NCS applies automatically, it is defined by the atoms in ncs_mask.pdb. All points within rad_mask (5 A in this example) of an atom in ncs_mask.pdb are considered to be within the region where NCS applies. This is useful if you want to apply NCS only to a part of the region where NCS is present.

Possible Problems

Specific limitations and problems:

Literature

Rapid automatic NCS identification using heavy-atom substructures. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2213-5 (2002).

- Rapid automatic NCS identification using heavy-atom substructures. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2213-5 (2002).

Statistical density modification with non-crystallographic symmetry. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2082-6 (2002).

- Statistical density modification with non-crystallographic symmetry. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 58, 2082-6 (2002).

Maximum-likelihood density modification. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 56, 965-72 (2000).

- Maximum-likelihood density modification. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 56, 965-72 (2000).

Map-likelihood phasing. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 57, 1763-75 (2001).

• Map-likelihood phasing. T.C. Terwilliger. Acta Crystallogr D Biol Crystallogr 57, 1763-75 (2001).

Additional information

List of all available keywords

input_filesdata_file = None Mtz or other data file with intensities or amplitudesdata_labels = None Optional data labels for Iobs,SIGIobs or Fobs,SIGFobs. Input data can be I, F, or I+/I- or F+/F-.phib_labels = None Optional data label for PHIB. You can force skipping with phib_labels=SKIPfom_labels = None Optional data label for FOM. You can force skipping with fom_labels=SKIPhl_file = None Mtz or other data file with HL coefficients. If None assumed to be the same as data_file. Required unless a pdb_file is supplied.hl_labels = None Optional labels for HL coefficients. You can force skipping with hl_labels=SKIPmap_coeffs_file = None Mtz or other file with map coefficients for starting map. If None assumed to be same as data_filemap_coeffs_labels = None Optional labels for map coefficients. You can force skipping with map_coeffs_labels=SKIPpdb_file = None Optional file with model to be used in density modification. If provided, the density modification will use model density as part of the density modification target. If supplied, HL coefficients are optional.ha_file = None Optional file with heavy-atom sites to be truncated in density modification. Density within rad_mask of a site in this file will be truncated at 2 x the rms of the map. Note: not compatible with mask_from_pdb.mask_from_pdb = None Optional file with dummy atoms marking the region to be considered as the macromolecule in density modification. The region within rad_mask of an atom in this file will be considered within the macromolecule. Note: not compatible with ha_file.add_mask = None Mask will be the union of the standard mask and mask_from_pdbncs_file = None Optional file with NCS information (.ncs_spec file). This file can be obtained with phenix.find_ncs or phenix.simple_ncs_from_pdbncs_domain_pdb = None Optional file with dummy atoms marking the region where NCS is to be applied. Supersedes ncs_domain_pdb in the ncs_file. Points within rad_mask of an atom in this file are considered for NCS. You need to supply a mask covering all NCS copies. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.resolve_commands_file = None Optional file with any commands for resolve density modification.directoriestemp_dir = temp_denmod Temporary directory. Not normally used. Contents deleted after the run if clean_up=True is specified.crystal_inforesolution = None Resolution for density modificationoutput_filesoutput_mtz = denmod.mtz Output MTZ file with density-modified map coefficientshl_in_output_mtz = True Calculate density modification HL coefficients and write to output mtz filetarget_map_file_name = target_map.ccp4 Output CCP4-style map with target map. only written if write_target_map_and_sigma is setsigma_map_file_name = sigma_map.ccp4 Output CCP4-style map with sigma map. only written if write_target_map_and_sigma is setoutput_mask_file = solvent_boundary.map Output CCP4-style map file with solvent boundary mask.output_ncs_mask_file = ncs_boundary.map Output CCP4-style map file with mask marking region used for NCS. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.anisotropyremove_aniso = True Remove anisotropy from data and optionally sharpenb_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)max_b_iso = 40. Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)target_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, ta...

- input_filesdata_file = None Mtz or other data file with intensities or amplitudesdata_labels = None Optional data labels for Iobs,SIGIobs or Fobs,SIGFobs. Input data can be I, F, or I+/I- or F+/F-.phib_labels = None Optional data label for PHIB. You can force skipping with phib_labels=SKIPfom_labels = None Optional data label for FOM. You can force skipping with fom_labels=SKIPhl_file = None Mtz or other data file with HL coefficients. If None assumed to be the same as data_file. Required unless a pdb_file is supplied.hl_labels = None Optional labels for HL coefficients. You can force skipping with hl_labels=SKIPmap_coeffs_file = None Mtz or other file with map coefficients for starting map. If None assumed to be same as data_filemap_coeffs_labels = None Optional labels for map coefficients. You can force skipping with map_coeffs_labels=SKIPpdb_file = None Optional file with model to be used in density modification. If provided, the density modification will use model density as part of the density modification target. If supplied, HL coefficients are optional.ha_file = None Optional file with heavy-atom sites to be truncated in density modification. Density within rad_mask of a site in this file will be truncated at 2 x the rms of the map. Note: not compatible with mask_from_pdb.mask_from_pdb = None Optional file with dummy atoms marking the region to be considered as the macromolecule in density modification. The region within rad_mask of an atom in this file will be considered within the macromolecule. Note: not compatible with ha_file.add_mask = None Mask will be the union of the standard mask and mask_from_pdbncs_file = None Optional file with NCS information (.ncs_spec file). This file can be obtained with phenix.find_ncs or phenix.simple_ncs_from_pdbncs_domain_pdb = None Optional file with dummy atoms marking the region where NCS is to be applied. Supersedes ncs_domain_pdb in the ncs_file. Points within rad_mask of an atom in this file are considered for NCS. You need to supply a mask covering all NCS copies. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.resolve_commands_file = None Optional file with any commands for resolve density modification.
- data_file = None Mtz or other data file with intensities or amplitudes
- data_labels = None Optional data labels for Iobs,SIGIobs or Fobs,SIGFobs. Input data can be I, F, or I+/I- or F+/F-.
- phib_labels = None Optional data label for PHIB. You can force skipping with phib_labels=SKIP
- fom_labels = None Optional data label for FOM. You can force skipping with fom_labels=SKIP
- hl_file = None Mtz or other data file with HL coefficients. If None assumed to be the same as data_file. Required unless a pdb_file is supplied.
- hl_labels = None Optional labels for HL coefficients. You can force skipping with hl_labels=SKIP
- map_coeffs_file = None Mtz or other file with map coefficients for starting map. If None assumed to be same as data_file
- map_coeffs_labels = None Optional labels for map coefficients. You can force skipping with map_coeffs_labels=SKIP
- pdb_file = None Optional file with model to be used in density modification. If provided, the density modification will use model density as part of the density modification target. If supplied, HL coefficients are optional.
- ha_file = None Optional file with heavy-atom sites to be truncated in density modification. Density within rad_mask of a site in this file will be truncated at 2 x the rms of the map. Note: not compatible with mask_from_pdb.
- mask_from_pdb = None Optional file with dummy atoms marking the region to be considered as the macromolecule in density modification. The region within rad_mask of an atom in this file will be considered within the macromolecule. Note: not compatible with ha_file.

• add_mask = None Mask will be the union of the standard mask and mask_from_pdb

• ncs_file = None Optional file with NCS information (.ncs_spec file). This file can be obtained with phenix.find_ncs or phenix.simple_ncs_from_pdb

• ncs_domain_pdb = None Optional file with dummy atoms marking the region where NCS is to be applied. Supersedes ncs_domain_pdb in the ncs_file. Points within rad_mask of an atom in this file are considered for NCS. You need to supply a mask covering all NCS copies. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.

• resolve_commands_file = None Optional file with any commands for resolve density modification.

• directoriestemp_dir = temp_denmod Temporary directory. Not normally used. Contents deleted after the run if clean_up=True is specified.

• temp_dir = temp_denmod Temporary directory. Not normally used. Contents deleted after the run if clean_up=True is specified.

• crystal_inforesolution = None Resolution for density modification

• resolution = None Resolution for density modification

• output_filesoutput_mtz = denmod.mtz Output MTZ file with density-modified map coefficientshl_in_output_mtz = True Calculate density modification HL coefficients and write to output mtz filetarget_map_file_name = target_map.ccp4 Output CCP4-style map with target map. only written if write_target_map_and_sigma is setsigma_map_file_name = sigma_map.ccp4 Output CCP4-style map with sigma map. only written if write_target_map_and_sigma is setoutput_mask_file = solvent_boundary.map Output CCP4-style map file with solvent boundary mask.output_ncs_mask_file = ncs_boundary.map Output CCP4-style map file with mask marking region used for NCS. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.

• output_mtz = denmod.mtz Output MTZ file with density-modified map coefficients

• hl_in_output_mtz = True Calculate density modification HL coefficients and write to output mtz file

• target_map_file_name = target_map.ccp4 Output CCP4-style map with target map. only written if write_target_map_and_sigma is set

• sigma_map_file_name = sigma_map.ccp4 Output CCP4-style map with sigma map. only written if write_target_map_and_sigma is set

• output_mask_file = solvent_boundary.map Output CCP4-style map file with solvent boundary mask.

• output_ncs_mask_file = ncs_boundary.map Output CCP4-style map file with mask marking region used for NCS. NOTE: the output_ncs_mask_file map shows the boundary of the asymmetric unit of NCS (just one copy) while the input ncs_domain_pdb file must contain all NCS copies.

• anisotropyremove_aniso = True Remove anisotropy from data and optionally sharpenb_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)max_b_iso = 40. Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)target_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)

• remove_aniso = True Remove anisotropy from data and optionally sharpen

- b_iso = None Target overall B value for anisotropy correction. Ignored if remove_aniso = False. If None, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- max_b_iso = 40. Default maximum overall B value for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- target_b_ratio = 10. Default ratio of target B value to resolution for anisotropy correction. Ignored if remove_aniso = False. Ignored if b_iso is set. If used, default is minimum of (max_b_iso, B of dataset, target_b_ratio*resolution)
- denmodrad_wang = None Radius for estimation of solvent boundary. Normally chosen automatically.optimize_rad_wang = False Optimize value of rad_wangmask_type = *None histograms probability wang Method for solvent identification. Default is histograms. If you have a SAD dataset with a heavy atom such as Pt or Au then you may wish to choose wang because the histogram method is sensitive to very high peaks. Options are: histograms: compare local rms of map and local skew of map to values from a model map and estimate probabilities. This one is usually the best. probability: compare local rms of map to distribution for all points in this map and estimate probabilities. In a few cases this one is much better than histograms. wang: take points with highest local rms and define as protein.solvent_content = None Solvent content (0 to 1) of the crystal (required)mask_cycles = 5 Mask cycles (overall cycles of density modification)minor_cycles = 10 Minor cycles of density modification for each mask cyclerad_mask = 4 Radius (A) for mask calculation around heavy-atom sites (ha_file), for definition of protein boundary (mask_from_pdb), and for definition of the region used for NCS (ncs_domain_pdb).denmod_with_model = None By default if a pdb_file is supplied it will be used in density modification. You can turn this off with denmod_with_model=Falseremove_waters = True Remove waters from model before use in density modification or mask generationremove_hetatm = True Remove HETATM (ligands, metals etc.) from model before use in density modification or mask generationmap_phasing_ab_average_weight = 0.1 Add in this fraction of average sigma when calculating weightsmap_phasing_ab_weighting = False Use weighting in map_phasing_ab output from sigmaxwrite_target_map_and_sigma = False Write out map files for target rho and sigmamap_phasing_ab = None Use map phasing to get estimates of structure factors as x,y and sigmax,sigmay
- rad_wang = None Radius for estimation of solvent boundary. Normally chosen automatically.
- optimize_rad_wang = False Optimize value of rad_wang
- mask_type = *None histograms probability wang Method for solvent identification. Default is histograms. If you have a SAD dataset with a heavy atom such as Pt or Au then you may wish to choose wang because the histogram method is sensitive to very high peaks. Options are: histograms: compare local rms of map and local skew of map to values from a model map and estimate probabilities. This one is usually the best. probability: compare local rms of map to distribution for all points in this map and estimate probabilities. In a few cases this one is much better than histograms. wang: take points with highest local rms and define as protein.
- solvent_content = None Solvent content (0 to 1) of the crystal (required)
- mask_cycles = 5 Mask cycles (overall cycles of density modification)
- minor_cycles = 10 Minor cycles of density modification for each mask cycle
- rad_mask = 4 Radius (A) for mask calculation around heavy-atom sites (ha_file), for definition of protein boundary (mask_from_pdb), and for definition of the region used for NCS (ncs_domain_pdb).
- denmod_with_model = None By default if a pdb_file is supplied it will be used in density modification. You can turn this off with denmod_with_model=False

- remove_waters = True Remove waters from model before use in density modification or mask generation
- remove_hetatm = True Remove HETATM (ligands, metals etc.) from model before use in density modification or mask generation
- map_phasing_ab_average_weight = 0.1 Add in this fraction of average sigma when calculating weights
- map_phasing_ab_weighting = False Use weighting in map_phasing_ab output from sigmax
- write_target_map_and_sigma = False Write out map files for target rho and sigma
- map_phasing_ab = None Use map phasing to get estimates of structure factors as x,y and sigmax,sigmay
- controln_xyz = None gridding for mapsverbose = False Verbose outputclean_up = True Remove temp_dir when finished
- n_xyz = None gridding for maps
- verbose = False Verbose output
- clean_up = True Remove temp_dir when finished